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Subject: Programming with Java	
ASSIGNMENT No : 2	
TITLE : Constructors in Java	

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Programme

```

1  class Addition
2  {
3      int a, b;
4      // Default
5      Addition()
6      {
7          a = 1;
8          b = 2;
9          System.out.println("Inside Default Constructor");
10     }
11
12     // Parameterized
13     Addition(int a, int b)
14     {
15         this.a = a;
16         this.b = b;
17         System.out.println("Inside Parameterized Constructor");
18     }
19
20     // Copy
21     Addition(Addition obj)
22     {
23         a = obj.a;
24         b = obj.b;
25         System.out.println("Inside Copy Constructor");
26     }
27
28     public static void main(String[] args)
29     {
30         Addition A1 = new Addition();
31         Addition A2 = new Addition(10, 20);
32
33         Addition A3 = new Addition(A2);
34     }
35
36

```

Output

```
piyushrohokale@Piyushs-MacBook-Air assignment02 % javac Addition.java && java Addition
Inside Default Constructor
Inside Parameterized Constructor
Inside Copy Constructor
```

Conclusion

The program successfully demonstrates the use of default, parameterized, and copy constructors in Java. The default constructor initializes objects with predefined values, the parameterized constructor initializes objects with user-defined values, and the copy constructor creates a new object by copying the data from an existing object. This shows how constructors are used for object initialization and efficient data management in Java.